Variable Delay with Stage Structure Write Up Summer 2026

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1 Introduction

The goal of this document is to introduce a variable delay into the stage-structured Daphnia system proposed by Ascoli et al. 2025, matching the variable-delay structure used in McCauley et al. 2008. All code used to generate figures is contained within our GITHUB Repository. Its important to mention that that Chat GPT was used to help write and test the majority of this code.

2 Ascoli et al. 2025 Model

Ascoli et al. propose the following stoichiometrically co-limited, stage-structured producer–grazer model. Let $x(t)$, $J(t)$, and $A(t)$ denote the carbon densities $\mathrm{mgC\,L^{-1}}$ of the producer, juvenile Daphnia, and adult Daphnia populations, respectively

$$\frac{dx}{dt} = bx(t)(1 - \frac{x(t)}{K})(1 - \frac{q}{Q(t)}) - f_j(t)J(t) - f_a(t)A(t), \tag{1a}$$

$$\frac{dJ}{dt} = e_a h_j(Q(t)f_a(t)A(t) - e_j h_a(Q(t))f_j(t)J(t) - \delta J(t), \tag{1b}$$

$$\frac{dA}{dt} = e_j h_a(Q(t))f_j(t)J(t) - \delta_A A(t), \tag{1c}$$

where

$$Q(t) = \frac{P - \theta_J J(t) - \theta_A A(t)}{x(t)},$$

$$f_i(t) = f_i(x(t)) = \frac{c_i x(t)}{a_i + x(t)}, \quad h_i(Q(t)) = \tanh(\frac{Q(t)}{\theta_i})$$

with $i \in \{a, j\}$ for juveniles and adults respectively. We have stage-specific parameters $c_i, a_i, e_i, \theta_i, \delta_i$ in addition to other parameters b, K, q, P as defined in Table 5. For later reference, we define the juvenile birth rate $B(t)$ and maturation rate $M(t)$ by

$$B(t) = e_a h_j(Q(t)f_a(t)A(t), \tag{2}$$
$$M(t) = e_j h_a(Q(t))f_j(t)J(t). \tag{3}$$

Table 1: Parameters in the Tomas et al. stoichiometric co-limitation model. Here, $i \in \{j, a\}$ denotes the juvenile or adult stage, respectively.

Parameter	**Description**	**Units**
c_i	Maximum ingestion rate for stage i	d^{-1}
a_i	Half-saturation constant for the ingestion rate of stage i	$\mathrm{mg\,C\,L^{-1}}$
e_i	Maximum production efficiency for stage i	-
θ_i	Constant phosphorus-to-carbon ratio for stage i	$\mathrm{mg\,P\,mg\,C^{-1}}$
δ_i	Per-capita death rate for stage i	d^{-1}
b	Maximum per-capita producer growth rate	d^{-1}
K	Light-dependent producer carrying capacity	$\mathrm{mg\,C\,L^{-1}}$
q	Minimum producer phosphorus-to-carbon ratio required for growth	$\mathrm{mg\,P\,mg\,C^{-1}}$
P	Total phosphorus concentration in the system	$\mathrm{mg\,P\,L^{-1}}$

Note that we have rewritten $f_i(x)$ and Q as time-dependent quantities because they depend on the time-dependent state variables. This is only a notational change and will simplify the introduction of delayed terms later. We also interpret the loss rates in Ascoli et al. specifically as death rates, since these rates will later determine survival through the juvenile stage.

3 McCauley et al. 2008 Model

McCauley et al. (2008) propose the following stage-structured consumer–resource model with a variable juvenile maturation delay. Let $x(t)$ denote resource carbon density, and let $J(t)$ and $A(t)$ denote juvenile and adult densities, respectively. Unlike the Tomas model, $J(t)$ and $A(t)$ are measured as numbers of individuals per unit volume rather than carbon biomass.

$$\frac{dx}{dt} = bx(t)\left(1 - \frac{x(t)}{K}\right) - \phi_j(t)J(t) - \phi_a(t)A(t), \tag{4a}$$
$$\frac{dJ}{dt} = \frac{\chi}{\gamma}\sigma_a\phi_a(t)A(t) - \frac{\chi}{\gamma}\sigma_a\phi_a(t-\tau(t))A(t-\tau(t))S(t)\frac{g(t)}{g(t-\tau(t))} - \delta_j(t)J(t), \tag{4b}$$
$$\frac{dA}{dt} = \frac{\chi}{\gamma}\sigma_a\phi_a(t-\tau(t))A(t-\tau(t))S(t)\frac{g(t)}{g(t-\tau(t))} - \delta_a(t)A(t). \tag{4c}$$

where

$$\phi_i(t) = \frac{I_i x(t)}{f_h + x(t)}, \tag{5a}$$

$$g(t) = \sigma_j \phi_j(t), \tag{5b}$$

$$\delta_i(t) = \frac{\mu_i}{\sigma_i \phi_i(t)}, \qquad i \in \{j, a\}, \tag{5c}$$

$$S(t) = \exp\{-\int_{t-\tau(t)}^{t} \delta_j(\xi)\, d\xi\}, \tag{5d}$$

$$w = \int_{t-\tau(t)}^{t} g(\xi)\, d\xi. \tag{5e}$$

We have the following stage dependent parameters σ_i, μ_i, I_i in addition to other parameters χ, γ, w and f_h as defined in Table 2. For later reference, define the juvenile recruitment rate as

$$R_j(t) = \frac{\chi}{\gamma} \sigma_a \phi_a(t) A(t),$$

and the juvenile maturation rate as

$$R_a(t) = R_j(t - \tau(t)) S(t) \frac{g(t)}{g(t - \tau(t))}.$$

Also note the eating rate used in McCauley

$$\phi_i(t) = \frac{I_i x(t)}{f_h + x_(t)},$$

has units of $\mathrm{mg\,C\,L^{-1}\,N^{-1}\,d^{-1}}$.

Table 2: Additional parameters in the McCauley et al. stage-structured variable-delay model. Here, $i \in \{j, a\}$ denotes the juvenile or adult stage, respectively, and N is individuals.

Parameter	**Description**	**Units**
σ_i	Proportion of ingested carbon utilized by stage i	-
μ_i	Stage-specific mortality scalar	$\mathrm{mg\,C\,L^{-1}\,N^{-1}\,d^{-2}}$
I_i	Maximum adult ingestion rate	$\mathrm{mg\,C\,L^{-1}\,N^{-1}\,d^{-1}}$
χ	Proportion of adult net production allocated to reproduction	-
γ	Carbon required to produce one offspring	$\mathrm{mg\,C\,L^{-1}\,N^{-1}}$
w	Carbon gain required for an individual to complete juvenile development	$\mathrm{mg\,C\,L^{-1}\,N^{-1}}$
f_h	Half saturation constant in functional response	$\mathrm{mg\,C\,L^{-1}}$

To facilitate comparison with the Tomas model, we write x, b, and K in place of McCauley et al.'s F, q, and k respectively. We also write $g(t)$ in place of their juvenile growth function $h(t)$, since h_i already denotes a stoichiometric limitation function in the Tomas model. We retain χ, γ, μ_i, and w in the notation of McCauley et al., since they have no independently named counterparts in the Tomas model.

4 Key Differences

In this section, I will outline the key differences between the models that we must account for when introducing a variable delay to Ascoli's system (1).

Difference 1. *Ascoli uses units of* $\mathrm{mg\,C\,L^{-1}}$ *for* J *and* A *while McCauley uses* $(\mathrm{individuals\,L^{-1}})$.

This is a major difference between the two systems which appears most importantly in two places. Firstly, the type II eating rate in McCauley,

$$\phi_i(t) = \frac{I_i x(t)}{f_h + x(t)}$$

has units of $(\mathrm{mg\,C\,individual^{-1}\,day^{-1}})$, while the eating rate in Ascoli

$$f_i = \frac{c_i x(t)}{a_i + x(t)},$$

which has units of day^{-1}. Second, the developmental threshold w, which implicitly defines the juvenile stage duration $\tau(t)$, has units of $(\mathrm{mg\,C\,individual^{-1}})$.

Difference 2. *Ascoli uses scaling terms* $e_i(-)$ *and* $h_i(Q(t))(-)$ *to scale ingested carbon* $f_i(t)$*, while McCauley uses* $\chi(-), \gamma(\mathrm{mg\,C\,L^{-1}\,individuals^{-1}}), \sigma_i(-)$ *to scale ingested carbon* $\phi_i(t)$.

In Ascoli's system (1), these parameters appear in the birth term $B(t)$ and maturation term $M(t)$. In McCauley's system (5), these parameters appear in the recruitment term $R_j(t)$, maturation term $R_a(t)$, development rate $g(t)$, and death rate $\delta_i(t)$. We will need to establish specific correspondences between these parameters when we develop our final model.

Difference 3. *Ascoli uses an instantaneous maturation term while McCauley treats maturation as a development completion term for juveniles born at* $t - \tau(t)$

In Ascoli's system (1), the maturation term

$$M(t) = e_j h_a(Q(t)) f_j(t) J(t)$$

is instantaneous and uses the $e_j h_a(Q(t))$ term to scale the ingested carbon $f_j(t)$. In McCauley's system (5), the maturation term

$$R_j(t - \tau(t)) S(t) \frac{g(t)}{g(t - \tau(t))}$$

depends on the recruitment rate

$$R_j(t) = \frac{\chi}{\gamma}\sigma_a\phi_a(t)A(t),$$

as well a through stage probability $S(t)$, which depends on a variable instantaneous death rate, and a scaler dependent on the development rate's $g(t)$ and $g(t-\tau(t))$.

Difference 4. *Ascoli uses a constant death rate while McCauley uses a variable death rate.*

In Ascoli's system (1), the per-capita death rate is a constant δ_i. In McCauley's system (5), the death rate is a function

$$\delta_i(t) = \frac{\mu_i}{\sigma_i\phi_i(t)}$$

which depends on the eating rate $\phi_i(t)$.

Difference 5. *Ascoli uses different half-saturation constants a_j, a_a for the juvenile and adult eating rates $f_j(t), f_a(t)$, while McCauley uses f_h for both half-saturation constants.*

Ascoli's system (1), has eating rates

$$f_j(t) = \frac{c_j x(t)}{a_j + x(t)} \quad f_a(t) = \frac{c_a x(t)}{a_a + x(t)},$$

while McCauley has eating rates

$$\phi_j(t) = \frac{I_j x(t)}{f_h + x(t)} \quad \phi_a(t) = \frac{I_a x(t)}{f_h + x(t)}$$

5 Model Construction

In this constructions, all terms will be defined as they are within the Ascoli system (1). For example, we will be using the eating rates $f_j(t)$ and $f_a(t)$ as defined in that model. But we will largely be following the structure outlined in McCauley system (5).

5.1 Stoichiometric co-limitation maturation equation

The key term we need to update within the Ascoli model is the maturation term, which in System (1) is the term

$$M(t) = e_j h_a(Q(t)) f_j(t) J(t)$$

which is present in (1b) and (1b). To derive a variable delay version, we will use the conveyor-belt analogy presented by W. S. C. Gurney and R. M. Nisbet and seen in Figure (1).

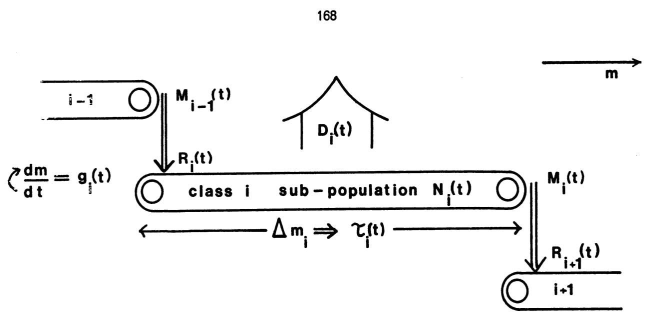


Figure 1: Caption

Firstly, let us define some terms. We will be taking the birth rate function

$$B(t) = e_a h_j(Q(t)) f_a(t) A(t)$$

directly from (1). $\rho_B(t)$ and $\rho_E(t)$ are the density of algae ($\mathrm{mg\,C\,L^{-1}}$) at the beginning and end of the juvenile conveyor belt at time t. $\hat{M}(t)$ is the rate of maturation, which we hope to solve for. Lastly, $d(t)$ is the development rate of the algae. Since we are accounting for the food quality of the producers as relating to the juveniles, we define

$$d(t) = e_j h_j(Q(t)) f_j(t)$$

.With these terms, we define the following equalities:

$$\hat{M}(t) = d(t) \cdot \rho_E(t) \tag{6a}$$

$$\rho_B(t) = \frac{B(t)}{d(t)} \tag{6b}$$

$$\rho_E(t) = \rho_B(t - \tau(t)) \cdot S(t) \tag{6c}$$

where $S(t)$ is the survival probability. This survival probability must be re re-derived; I will leave it for now. Now let us plug equation (6c) into equation (6a) yielding

$$\hat{M}(t) = d(t) \cdot \rho_B(t - \tau(t)) \cdot S(t)$$

and then plug $t - \tau(t)$ into equation (6b) which yields our final form

$$\hat{M}(t) = B(t - \tau(t)) \cdot \frac{d(t)}{d(t - \tau(t))} \cdot S(t) \tag{7}$$

$$= B(t - \tau(t)) \cdot \frac{h_j(Q(t)) \cdot f_j(t)}{h_j(Q(t - \tau(t))) \cdot f_j(t - \tau(t))} \cdot S(t) \tag{8}$$

5.2 Stoichiometric Survival Probability and Death Rate

Let us now define the Stoichiometric Survival Probability. As defined in (5),

$$S(t) = \exp\{-\int_{t-\tau(t)}^{t} \delta_j(\xi) d\xi\}$$

where $\delta_j(\xi) = \frac{\mu_j}{\sigma_j f_j(t)}$ is the variable death rate. We must first define a stoichiometric death rate. We have two options for this term, either we can match Ascoli and use a constant death rate, or we could match McCauley and use a variable death rate dependent on ingested carbon and P:C ratio of producers.

For a constant death rate of δ_j and δ_a, the survival probability is quite simple and can be reduced to an exponential term as follows:

$$S^*(t) = \exp\{-\int_{t-\tau(t)}^{t} \delta_j d\xi\} \tag{9}$$

$$= e^{-\delta_J \tau(t)} \tag{10}$$

For a variable death rate, we will need to introduce a new unitless mortality scaler $\hat{\mu_i}$, and can define our variable death rate as

$$\hat{\delta_i}(t) = \frac{\hat{\mu_i}}{e_i h_i(Q(t)) \cdot f_i(t)}.$$

Therefore, our stoichiometric survival probability is

$$\hat{S}(t) = \exp\{-\int_{t-\tau(t)}^{t} \hat{\delta_j}(\xi) d\xi\} \tag{11}$$

$$= \exp\{-\int_{t-\tau(t)}^{t} \frac{\hat{\mu_j}}{e_j h_j(Q(\xi)) \cdot f_j(\xi)} d\xi\} \tag{12}$$

5.3 Stage Duration

Let us now define our stage duration term $\hat{\tau(t)}$, which must depend on the amount of ingested carbon functions while factoring in the P:C ratio of the producer. However, we cannot use the same development parameter w since it has units of (mg C individual^{-1}). Thus, we will define a new unitless parameter $\hat{w}$ which implicitly defines our new stage duration term. Thus, we follow the structure of McCauley and use our development rate $d(t) = e_j h_j(Q(t)) f_j(t)$ and thus,

$$\hat{w} = \int_{t-\hat{\tau}(t)}^{t} e_j h_j(Q(\xi)) f_j(\xi) d\xi \tag{13a}$$

6 Model Presentation

Now let us construct our full system of equations, which uses a variable delay, stoichiometric co-limitation, and a constant death rate. First, let us present our parameter definitions in Table 3 and helper functions in Table 4. Note that from now on we will drop the hats on $\hat{M}(t), \hat{S(t)}, \hat{w}$.

Table 3: Parameters for the stage-structured stoichiometric co-limitation model.

Parameter	**Definition**	**Units**
b	Maximum per-capita rate of resource growth	d^{-1}
K	Resource carrying capacity	C
P	Total phosphorus in system	P
e_j	Juvenile max production efficiency	–
e_a	Adult max production efficiency	–
c_j	Maximum juvenile ingestion rate	d^{-1}
c_a	Maximum adult ingestion rate	d^{-1}
a_h	Half-saturation constant in the functional response	C
δ_j	Death rate of juveniles	d^{-1}
δ_a	Death rate of adults	d^{-1}
θ_j	Juvenile constant P:C ratio	$\mathrm{P\,C}^{-1}$
θ_a	Adult constant P:C ratio	$\mathrm{P\,C}^{-1}$
q	Producer min P:C	$\mathrm{P\,C}^{-1}$
w	Dimensionless developmental threshold required for juvenile maturation	–

Here, C denotes $\mathrm{mgC\,L}^{-1}$, P denotes $\mathrm{mgP\,L}^{-1}$, and d denotes days.

Table 4: Functions in the stage-structured stoichiometric co-limitation model.

Functions	
$g(t) = bx(t)(1 - \frac{x(t)}{K})(1 - \frac{q}{Q(t)})$	Resource growth function
$Q(t) = \frac{P - \theta_j J(t) - \theta_a A(t)}{x(t)}$	Cell quota function
$d(t) = e_j h_j(Q(t)) f_j(t)$	Juvenile development rate
$h_j(Q(t)) = \tanh(\frac{Q(t)}{\theta_j})$	Juvenile hyperbolic tangent function
$h_a(Q(t)) = \tanh(\frac{Q(t)}{\theta_a})$	Adult hyperbolic tangent function
$f_j(t) = \frac{c_j x(t)}{x(t) + a_h}$	Juvenile ingestion rate
$f_a(t) = \frac{c_a x(t)}{x(t) + a_h}$	Adult ingestion rate
$S(t) = e^{-\delta_j \tau(t)}$	Survival probability with constant death

Thus our balance equations are

$$\frac{dx}{dt} = \underbrace{g(t)}_{\text{logistic co-limitation growth}} - \underbrace{f_j(t)J(t)}_{\text{ingestion by juveniles}} - \underbrace{f_a(t)A(t)}_{\text{ingestion by adults}} \tag{14a}$$

$$\frac{dJ}{dt} = \underbrace{B(t)}_{\text{birth rate}} - \underbrace{M(t)}_{\text{maturation rate}} - \underbrace{\delta_j J(t)}_{\text{juvenile mortality}} \tag{14b}$$

$$\frac{dA}{dt} = \underbrace{M(t)}_{\text{maturation rate}} - \underbrace{\delta_a A(t)}_{\text{adult mortality}} \tag{14c}$$

and our maturation and birth rates are

$$B(t) = e_a h_j(Q(t)) f_a(t) A(t) \tag{15a}$$

$$M(t) = B(t - \tau(t)) \cdot \frac{d(t)}{d(t - \tau(t))} \cdot S(t) \tag{15b}$$

$$= B(t - \tau(t)) \cdot \frac{h_j(Q(t)) \cdot f_j(t)}{h_j(Q(t - \tau(t))) \cdot f_j(t - \tau(t))} \cdot S(t) \tag{15c}$$

with our through-stage survival probability being

$$S(t) = e^{-\delta_j \tau(t)} \tag{16}$$

and our stage duration defined as

$$w = \int_{t-\tau(t)}^{t} e_j h_j(Q(\xi)) f_j(\xi) d\xi \tag{17a}$$

7 Non-Dimensional Model

We now propose the following nondimensional model. Note that the juvenile and adult equations have been simplified by combining the birth and maturation terms.

Our nondimensional variables are

$$\widetilde{x} = \frac{x}{K}, \qquad \widetilde{J} = \frac{c_j J}{bK}, \qquad \widetilde{A} = \frac{c_a A}{bK}, \qquad \widetilde{Q} = \frac{Q}{q}, \qquad \widetilde{t} = bt, \qquad \widetilde{\tau} = b\tau.$$

The additional dimensionless parameters are

$$\lambda = \frac{a_h}{K}, \qquad \kappa_j = \frac{\theta_j b}{c_j q}, \qquad \kappa_a = \frac{\theta_a b}{c_a q}, \qquad \Pi = \frac{P}{qK}, \qquad r_j = \frac{q}{\theta_j},$$

$$\alpha = \frac{c_j e_a}{b}, \qquad \beta = \frac{c_a e_a}{b}, \qquad \chi_j = \frac{\delta_j}{b}, \qquad \chi_a = \frac{\delta_a}{b}, \qquad \rho = \frac{e_j c_j}{wb}.$$

The dimensionless functions are

$$\widetilde{Q}(\widetilde{t}) = \frac{\Pi - \kappa_j \widetilde{J}(\widetilde{t}) - \kappa_a \widetilde{A}(\widetilde{t})}{\widetilde{x}(\widetilde{t})},$$

$$\widehat{h}_j(\widetilde{Q}(\widetilde{t})) = \tanh(r_j \widetilde{Q}(\widetilde{t})), \qquad F(\widetilde{t}) = \frac{\widetilde{x}(\widetilde{t})}{\lambda + \widetilde{x}(\widetilde{t})},$$

$$D(\widetilde{t}) = \widehat{h}_j(\widetilde{Q}(\widetilde{t})) \cdot F(\widetilde{t}).$$

Dropping the tildes and the hat on $\widehat{h}_j$, we obtain

$$\frac{dx}{dt} = x(t)\Big(1 - x(t)\Big)\left(1 - \frac{1}{Q(t)}\right) - F(t)\big(J(t) + A(t)\big). \tag{18a}$$

$$\frac{dJ}{dt} = \alpha D(t)\left[A(t) - A(t-\tau(t))S(t)\right] - \chi_j J(t), \tag{18b}$$

$$\frac{dA}{dt} = \beta D(t) A(t-\tau(t)) S(t) - \chi_a A(t), \tag{18c}$$

$$1 = \rho \int_{t-\tau(t)}^{t} D(\xi)\, d\xi, \tag{18d}$$

$$S(t) = \exp\left[-\chi_j \tau(t)\right]. \tag{18e}$$

8 Change of Variables

Now we will perform a change of variables, from t to ϕ, using the following relations

$$\phi(t) = \rho \int_0^t D(\xi)\, d\xi, \qquad \frac{d\phi(t)}{dt} = \rho D(\phi) \tag{19}$$

Note that we can also invert this relation, yielding

$$\frac{dt}{d\phi} = \frac{1}{\rho D(\phi)} \tag{20}$$

From (19), we can write $\phi(t - \tau(t)) = \phi(t) - 1$, and thus on ingested carbon scale ϕ, our system can be written as

$$\frac{dx(\phi)}{d\phi} = \frac{x(\phi)}{\rho D(\phi)} (1 - x(\phi)) \left(1 - \frac{1}{Q(\phi)}\right) - \frac{J(\phi) + A(\phi)}{\rho h_j(Q(\phi))} \tag{21a}$$

$$\frac{dJ(\phi)}{d\phi} = \frac{\alpha}{\rho} [A(\phi) - A(\phi - 1)S(\phi)] - \frac{\chi_j}{\rho D(\phi)} J(\phi) \tag{21b}$$

$$\frac{dA(\phi)}{d\phi} = \frac{\beta}{\rho} A(\phi - 1)S(\phi) - \frac{\chi_a}{\rho D(\phi)} A(\phi) \tag{21c}$$

$$\frac{dt(\phi)}{d\phi} = \frac{1}{\rho D(\phi)} \tag{21d}$$

$$S(\phi) = \exp\left[-\chi_j (t(\phi) - t(\phi - 1))\right] \tag{21e}$$

9 Preliminary Figures

We now use the transformed model (21) to produce several preliminary numerical figures. Unless otherwise stated, we use the parameter values from Ascoli listed in Table 5.

Table 5: Baseline parameter values used in the preliminary numerical simulations.

Parameter	Definition	Baseline	Units
P	Total phosphorus in the system	0.025	$\mathrm{mg\,P\,L^{-1}}$
b	Maximum producer growth rate	1.2	$\mathrm{d^{-1}}$
q	Minimum producer P:C ratio	0.0038	$\mathrm{mg\,P\,mg^{-1}\,C}$
e_j	Maximum juvenile production efficiency	0.5	–
e_a	Maximum adult production efficiency	0.8	–
θ_j	Constant juvenile P:C ratio	0.025	$\mathrm{mg\,P\,mg^{-1}\,C}$
θ_a	Constant adult P:C ratio	0.03	$\mathrm{mg\,P\,mg^{-1}\,C}$
δ_j	Juvenile loss rate	0.06	$\mathrm{d^{-1}}$
δ_a	Adult loss rate	0.08	$\mathrm{d^{-1}}$
a_h	Juvenile and adult ingestion half-saturation constant	0.25	$\mathrm{mg\,C\,L^{-1}}$
c_j	Maximum juvenile ingestion rate	0.5	$\mathrm{d^{-1}}$
c_a	Maximum adult ingestion rate	0.81	$\mathrm{d^{-1}}$

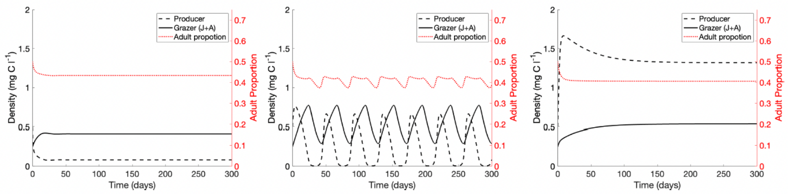
The MATLAB code used to generate the following figures is available in the `combined_model_simulations_varying_K.m`.

Because an estimated value of the developmental threshold w is not currently available, we set $w = 1$ for these preliminary simulations. This choice is arbitrary, and the value of w can substantially affect the resulting dynamics. Additionally, we must choose history functions since we are now working with delayed equations. For $\frac{dx}{d\phi}$, $\frac{dJ}{d\phi}$, and $\frac{dA}{d\phi}$ we chose constant history functions mirroring the initial conditions chosen by Ascoli of $x_0 = 0.5, J_0 = 0.125, A_0 = 0.125$. Since we also have a $\frac{dt}{d\phi}$ term which is used in our $S(\phi)$ with a delay, we also must specify a history function for our time term. For the following figures, we used a linear history function where $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$. This can be seen in our code in the "History Function" block near the top of the file.

9.1 Population Dynamics

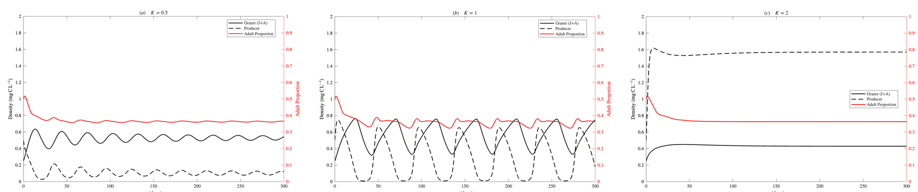
We first recreate the population simulations presented in Figure 2 of Ascoli. The original figure, including its caption, is reproduced below for comparison.

The corresponding simulations produced by the combined model are shown in Figure 2.



(a) $K = 0.5$ mg C/l (b) $K = 1$ mg C/l (c) $K = 2$ mg C/l

Figure 2: Numerical simulation of the model (5) showing population densities (black) and the proportion of C in the grazer population composed of adults (red), for parameter values listed Table 1 and varying values for K representing **(a)** low light levels $K = 0.5$ mg C/l, **(b)** medium light levels $K = 1$ mg C/l, and **(c)** high light levels $K = 2$ mg C/l. Initial conditions are $x(0) = 0.5$ mg C/l, $J(0) = 0.125$ mg C/l, and $A(0) = 0.125$ mg C/l.



(a) $K = 0.5 \text{ mg C L}^{-1}$ (b) $K = 1 \text{ mg C L}^{-1}$ (c) $K = 2 \text{ mg C L}^{-1}$

Figure 2: Population dynamics produced by the combined model for three values of the producer carrying capacity K.

9.2 Producer P:C Ratio

We next recreate Figure 3 of Ascoli, which shows the variable producer P:C ratio $Q(t)$ for different values of K. The original figure and its caption are reproduced below.

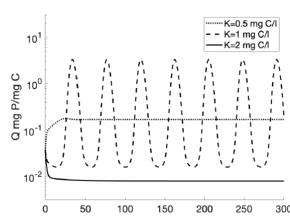


Figure 3: Numerical simulation of the model (5) showing Q, the variable P:C ratio of the producer, for parameter values listed Table 1 and varying values for K. Initial conditions are $x(0) = 0.5$ mg C/l, $J(0) = 0.125$ mg C/l, and $A(0) = 0.125$ mg C/l. Simulations correspond with those in Fig. 2.

Our corresponding simulation is shown in Figure 3.

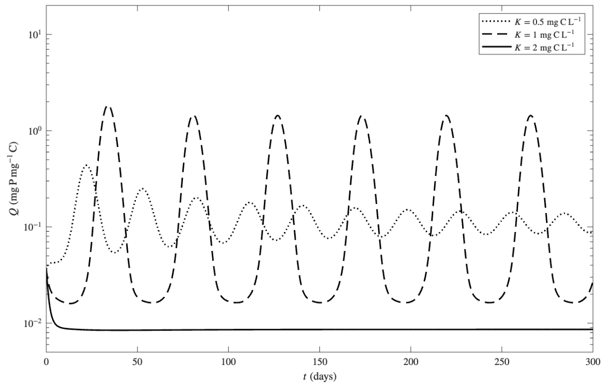


Figure 3: Producer P:C ratio $Q(t)$ produced by the combined model for three values of K.

9.3 Variable Stage Duration

Finally, we recreate the stage-duration plots from Figure S1 of McCauley. These plots compare the variable juvenile stage duration, shown by the solid line, with the estimated population-cycle period, shown by the dashed line. In the original figure, the two simulations use $K = 0.6$ and $K = 0.3$, respectively. The original plots and their caption are reproduced below.

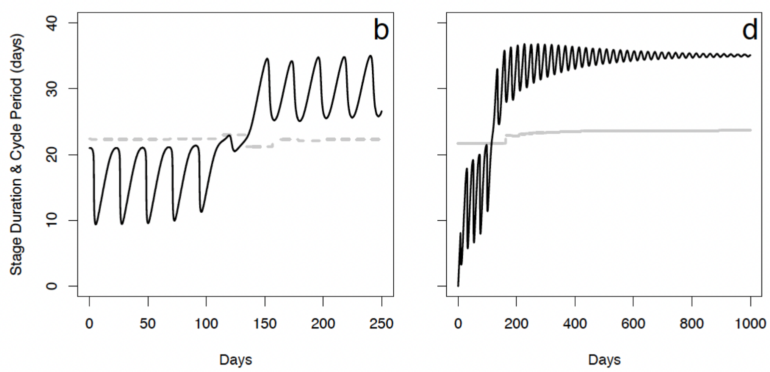
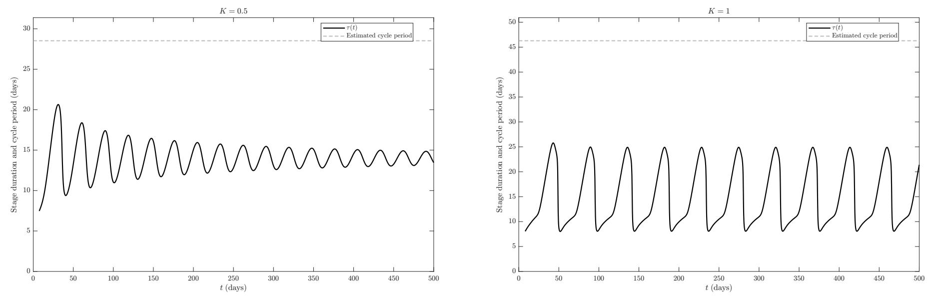


Figure 4 shows the corresponding stage-duration simulations produced by the combined model for two values of K.



(a) $K = 0.5$ mg C L^{-1} (b) $K = 1$ mg C L^{-1}

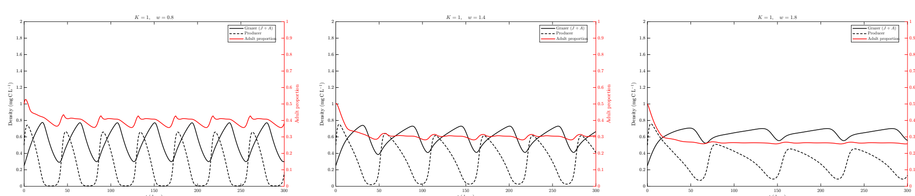
Figure 4: Variable juvenile stage duration and estimated population-cycle period produced by the combined model for two values of K.

10 Additional Figures

In this section, we will present additional preliminary figures and comment on the types of behavior we observe.

10.1 Varied w results

The code used for this section is in the GITHUB repository linked above under `combined_model_simulations_varying_w.m`. In Figure 2, we showcase time-series results for a set of $w = 1$ values and varied carrying capacities K. In our combined model, w is a yet-to-be-found dimensionless parameter, and thus we present time series simulations for a fixed carrying capacity of $K = 1$ and varied w values in Figure 5. For the figures in this section, we use a constant history function of $x(s) = 0.5, J(s) = 0.125, A(s) = 0.125$ and a linear history function $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$.



(a) $w = 0.8$ (b) $w = 1.4$ (c) $w = 1.8$

Figure 5: Population dynamics produced by the combined model for three values of the developmental threshold w with a set carrying capacity of $K = 1$.

We can see that for a K value that yields cycles, a higher w seems to dampen the oscillation amplitude. We also plot the variable P:C ratio $Q(t)$ for different values of w in Figure 6.

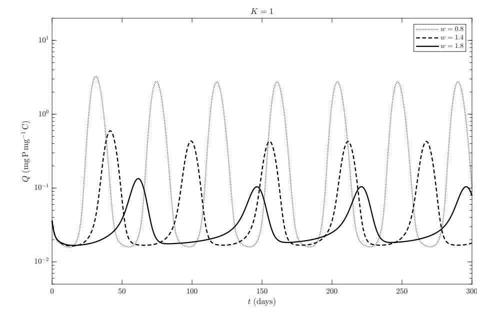
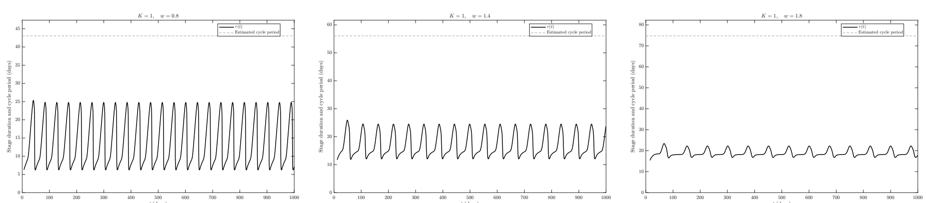


Figure 6: Producer P:C ratio $Q(t)$ produced by the combined model for three values of w and a set carrying capacity of $K = 1$.

We observe a reduction in amplitude and a lengthening of the period as w increases. Lastly, let us plot the stage duration and cycle period for our varied w values in Figure 7.



(a) $w = 0.8$ (b) $w = 1.4$ (c) $w = 1.8$

Figure 7: Population dynamics produced by the combined model for three values of the developmental threshold w with a set carrying capacity of $K = 1$.

Note that the amplitude of the variation in stage duration decreases, but the average stage duration remains around 20 days for all three w values. Also note that the average period length increases for higher values of w.

10.2 Simple Bifurcation Diagrams

The code used in this section is in the same GITHUB repository linked above under `bifurcation_single_w.m` and `bifurcation_multiple_w.m`.

A key feature of the stoichiometric model proposed by Tomas is the bifurcation structure of the grazers and producers as carrying capacity increases. In Tomas's model, as K increases, we observe that an interior equilibrium gains stability, and then, through a Hopf bifurcation, a limit cycle emerges. The oscillation amplitudes keep increasing before an eventual collapse at a saddle-node bifurcation into a new stable interior equilibrium. It will be a good sanity check to confirm that our combined model maintains this behavior. Note that for all figures in this section, We use a constant history function of $x(s) = 0.3125, J(s) = 0.09, A(s) = 0.01$ and a linear history function $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$.

Thus, we first present a single bifurcation plot that captures the maximum and minimum behavior in the long term for an increasing K value and a fixed w value in Figure 8

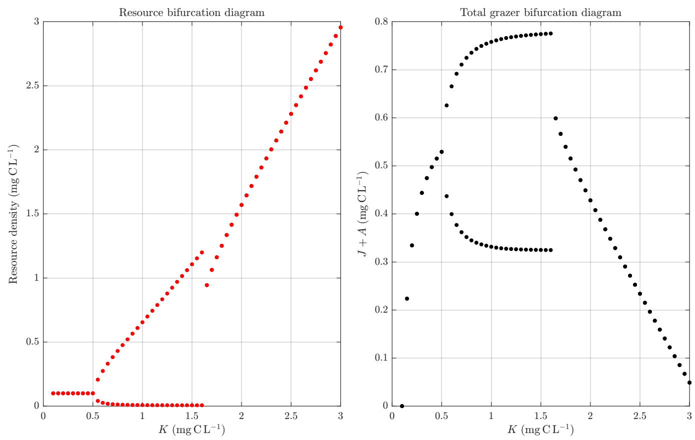


Figure 8: Bifurcation diagram of combined model with a fixed $w = 1$

In the above plot, we see the same behavior as in Tomas's model. Next, we present another bifurcation plot in Figure 9 showing multiple values of w. Additionally, we track the oscillation period when we are in a limit cycle.

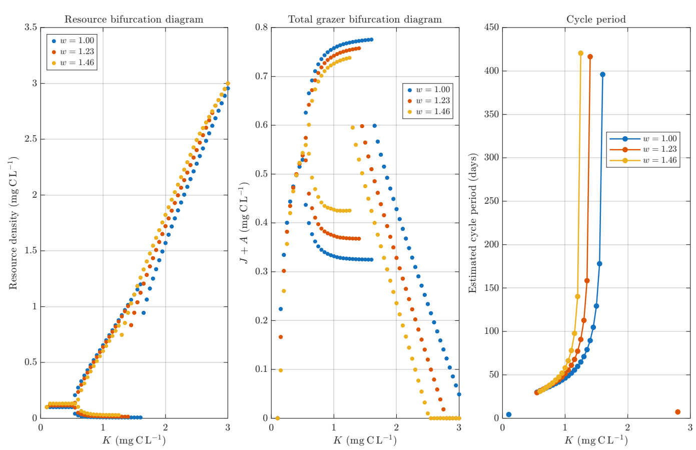


Figure 9: Bifurcation diagram of combined model with different w values as well as a plot of the cycle periods

Notice in the bifurcation plots that higher values of w result in smaller amplitudes during the limit cycles and a quicker collapse to an interior equilibrium.

In the cycle period plot, the periods increase rapidly as K approaches the point at which the limit cycles collapse.

10.3 Heatmaps

In this section, we present four heatmaps illustrating model behavior as K and w vary. The code used to make the matrices and then plot them in this section is in the GITHUB repository under `combined_model_heatmaps.m` and `plot_combined_model_heatmaps.m`, as well as the actual data from this run under `combined_model_heatmaps_results.mat`. Note that for all figures in this section, that we use a constant history function of $x(s) = 0.3125, J(s) = 0.125, A(s) = 0.125$ and a linear history function $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$.

First, Figure 10 shows the amplitudes of our grazer limit cycles. Blue represents a collapse to a stable interior equilibrium (and thus an amplitude of zero), while white represents extinction.

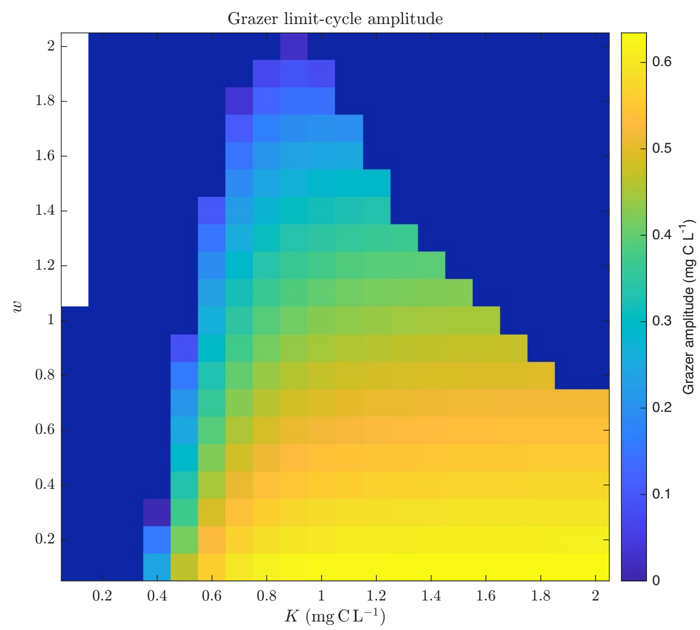


Figure 10: Two-parameter bifurcation diagram showcasing the limit cycle amplitude of our combined model as K and w vary.

Note that the amplitude appears to generally increase for higher carrying capacities and generally decrease for higher w values.

Next, Figure 11 shows the cycle period of our grazer limit cycles. Again, blue represents a collapse to a stable interior equilibrium, while white represents extinction.

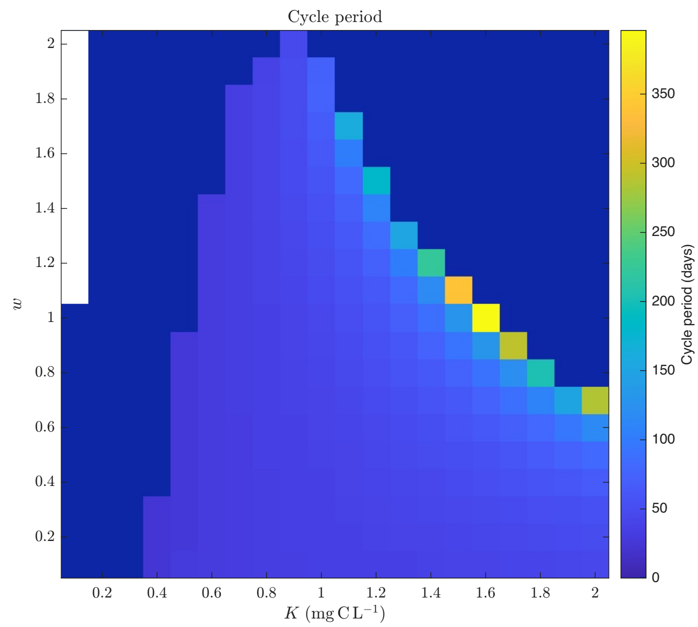


Figure 11: Two-parameter bifurcation diagram showcasing the limit cycle period of our combined model as K and w vary

Notice that along the upper-right boundary of the limit cycle parameter combinations, we begin to see exceedingly long periods that are not biologically relevant. This behavior was also shown in Figure 9. Also note that larger w tends to result in longer periods as K increases.

Next, Figure 12 shows the average stage duration $\tau(t)$ of our juvenile grazers. As seen previously, $\tau(t)$ oscillates, so this plot simply takes the long-term average. Another useful metric would be plotting the amplitude of the oscillations, but I did not think of that before running these simulations.

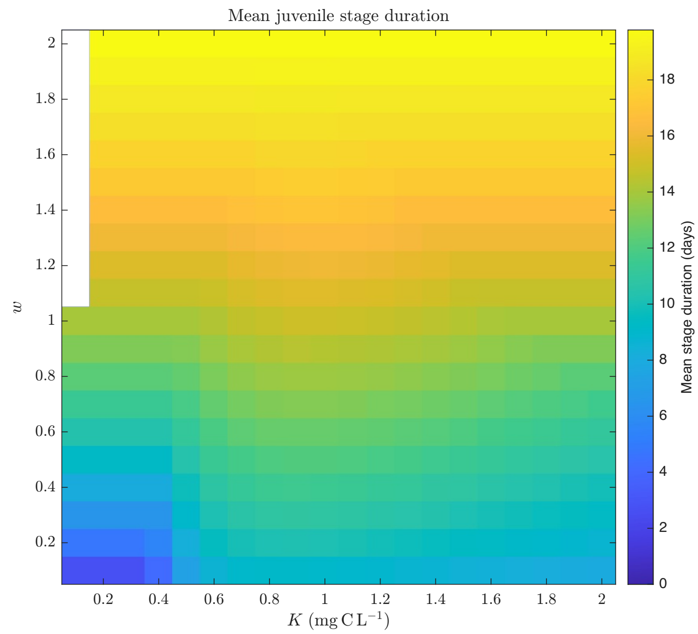
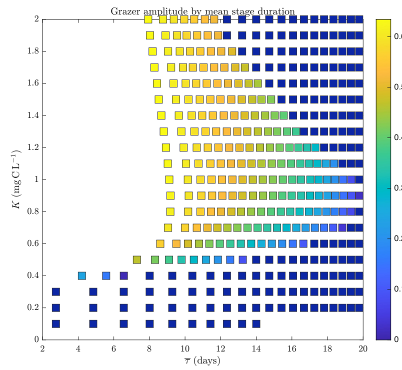


Figure 12: Two-parameter bifurcation diagram showcasing the average juvenile stage duration limit of our combined model as K and w vary

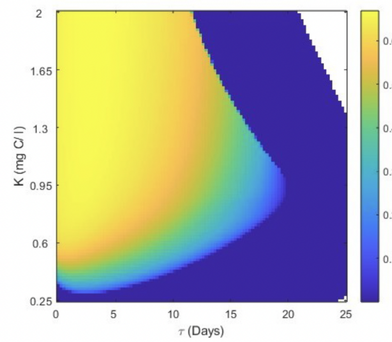
The behavior of this plot suggests that carrying capacity has little effect on the mean juvenile stage duration, and that larger w leads to a longer stage duration (which makes a lot of sense). Comparing this plot to Figure 11, we can see[1] that for every K and w combination, the average stage duration is less than the grazer period. This indicates that we will not observe any "large-amplitude vs small-amplitude" limit cycles, as in the McCauley system.

Lastly, we present a slightly different heatmap for comparison with a figure in Ricardo's fixed-delay paper. In Figure 13, on the left, we present a heatmap of average stage duration on the x-axis and K on the y-axis, with the color indicating amplitude of grazer oscillations. On the right, we have Ricardo's figure with fixed stage duration on the x-axis.

[1]the figure has a globally relevant scale, but upon changing the scale the minimum period is around 20 days with the corresponding stage duration being 2-4 days



(a)



(b)

Figure 13: Bifurcation plots showcasing the long-term relationship of stage duration (average on the left, constant on the right) and K on grazer amplitude.

In comparing Figure 13a to Figure 13b, we can see a similar shape and apparent pattern. We most likely expect this pattern, but maybe this indicates that a variable delay, so far, adds little while complicating our analysis.

As a final comment, the effect of higher w values seem to be lowering the probability of through-stage survival. Thus, juveniles do not survive maturation leading to either extinction or interior equilibrium without cycles.

10.4 Comments on Combined Model

In this section, we will outline some key behaviors present, and more importantly, not present in our combined model as compared to McCauley and Ascoli's systems.

First, let us discuss similarities, which largely come from Ascoli's system. The bifurcation diagrams seen in section 10.2, showcase the same interior equilibrium to limit-cycle back to interior equilibrium behavior seen in LKE and Ascoli. We also see similar time-series population dynamics results as in Ascoli in section 9.1.

Now, let us discuss differences between our combined model and the McCauley model. First, in our simulations we are unable to yield a longer stage duration $\tau(t)$ then grazer oscillation period. This relationship is important because McCauley identified it as a key feature distinguishing the small- and large-amplitude limit cycles. We also found no evidence of coexisting limit cycles with different amplitudes. In addition, our simulations did not display the abrupt change in stage duration shown in the Nature supplement and in the figures reproduced in our McCauley write-up.

Since the large and small limit cycles seems like a key behavior seen in observable data, the above differences feel like a significant issue with our combined model. The reason for this difference might be hinted at by McCauley who noted that "removing the type II functional response that is the source of the instability in nutrient-rich algal environments, decoupling the prey dynamics by replacing density-dependent growth with semi-chemostat dynamics, fixing the juvenile stage duration at time periods longer than the cycle period, all cause the coexisting attractors to disappear." Focusing on that first sentence, we believe that the variable mortality rate dependent on a type II functional response is a key mechanism of producing our desired behavior. This motivates the variable-mortality combined model introduced in the following section.

It is also important to note that initial conditions and history functions play a large role in the behaviors of the McCauley model, and although we did test various initial conditions, it is possible we did not find the right histories which yielded our desired behavior.

11 Variable-Mortality Combined Model

In this section and the results that follow, we define a new combined model that uses development-dependent mortality rates and the corresponding through-stage survival probability. The mortality rates and survival probability follow the formulation introduced in Section 5.2. We introduce this model to determine whether resource-dependent mortality is necessary for producing coexisting large- and small-amplitude limit cycles.

We retain the same state variables, biological parameters, dimensionless parameters, and helper functions as in the previous combined model, except where additions or changes are stated below.

11.1 Parameters and Helper Functions

The additional dimensional parameters introduced by the variable-mortality formulation are listed in Table 6. The remaining dimensional parameters are unchanged from the constant-mortality model.

Table 6: Additional parameters for the variable-mortality combined model.

Parameter	Definition	Units
μ_j	Juvenile mortality scaling parameter	d^{-2}
μ_a	Adult mortality scaling parameter	d^{-2}
r_a	Dimensionless ingestion scaler $r_a = \dfrac{q}{\theta_a}$	–
χ_j	Dimensionless juvenile mortality parameter, $\chi_j = \dfrac{\mu_j}{e_j c_j b}$	–
χ_a	Dimensionless adult mortality parameter, $\chi_a = \dfrac{\mu_a}{e_a c_a b}$	–

The units of μ_i follow from the requirement that $\delta_i(t)$ have units of d^{-1}. In particular, e_i and h_i are dimensionless, while $f_i(t)$ has units of d^{-1}. Therefore, μ_i must have units of d^{-2}.

The dimensional helper functions used in the model are given in Table 7.

Table 7: Helper functions for the variable-mortality combined model.

Functions	
$g(t) = bx(t)\left(1 - \frac{x(t)}{K}\right)\left(1 - \frac{q}{Q(t)}\right)$	Stoichiometrically limited resource growth
$Q(t) = \frac{P - \theta_j J(t) - \theta_a A(t)}{x(t)}$	Resource phosphorus-to-carbon ratio
$h_i(Q(t)) = \tanh\left(\frac{Q(t)}{\theta_i}\right), \quad i \in \{j, a\}$	Stoichiometric production-limitation function
$f_i(t) = \frac{c_i x(t)}{a_h + x(t)}, \quad i \in \{j, a\}$	Stage-specific ingestion rate
$d(t) = e_j h_j(Q(t)) f_j(t)$	Juvenile developmental rate
$\delta_i(t) = \frac{\mu_i}{e_i h_i(Q(t)) f_i(t)}, \quad i \in \{j, a\}$	Stage-specific variable mortality rate
$B(t) = e_a h_j(Q(t)) f_a(t) A(t)$	Production rate of juvenile carbon biomass
$S(t) = \exp\left\{-\int_{t-\tau(t)}^{t} \delta_j(\xi)\, d\xi\right\}$	Juvenile through-stage survival probability

11.2 Dimensional Model

The dimensional balance equations are

$$\frac{dx}{dt} = g(t) - f_j(t)J(t) - f_a(t)A(t), \tag{22a}$$
$$\frac{dJ}{dt} = B(t) - M(t) - \delta_j(t)J(t), \tag{22b}$$
$$\frac{dA}{dt} = M(t) - \delta_a(t)A(t). \tag{22c}$$

As before, the variable juvenile stage duration is defined implicitly by

$$w = \int_{t-\tau(t)}^{t} e_j h_j(Q(\xi)) f_j(\xi)\, d\xi. \tag{23}$$

The maturation rate is

$$M(t) = B(t-\tau(t))\frac{d(t)}{d(t-\tau(t))}S(t) \tag{24}$$
$$= e_a h_j(Q(t)) f_a(t) A(t-\tau(t)) S(t).. \tag{25}$$

The juvenile survival probability is

$$S(t) = \exp\left\{-\int_{t-\tau(t)}^{t} \frac{\mu_j}{e_j h_j(Q(\xi)) f_j(\xi)}\, d\xi\right\}. \tag{26}$$

11.3 Non-Dimensional Model

We use the same nondimensional variables as in the constant-mortality model:

$$\tilde{x} = \frac{x}{K}, \qquad \tilde{J} = \frac{c_j J}{bK}, \qquad \tilde{A} = \frac{c_a A}{bK}, \qquad \tilde{Q} = \frac{Q}{q}, \qquad \tilde{t} = bt, \qquad \tilde{\tau} = b\tau. \tag{27}$$

We also retain the dimensionless parameters

$$\lambda = \frac{a_h}{K}, \qquad \kappa_j = \frac{\theta_j b}{c_j q}, \qquad \kappa_a = \frac{\theta_a b}{c_a q}, \qquad \Pi = \frac{P}{qK}, \tag{28}$$
$$r_j = \frac{q}{\theta_j}, \qquad \alpha = \frac{c_j e_a}{b}, \qquad \beta = \frac{c_a e_a}{b}, \qquad \rho = \frac{e_j c_j}{wb}. \tag{29}$$

The new dimensionless mortality and ingestion parameters are

$$\chi_j = \frac{\mu_j}{e_j c_j b}, \qquad \chi_a = \frac{\mu_a}{e_a c_a b}, \qquad r_a = \frac{q}{\theta_a}. \tag{30}$$

Define

$$Q(t) = \frac{\Pi - \kappa_j J(t) - \kappa_a A(t)}{x(t)}, \qquad F(t) = \frac{x(t)}{\lambda + x(t)}, \tag{31a}$$
$$\tilde{h}_i(Q(t)) = \tanh\left(r_i Q(t)\right), \qquad D_i(t) = \tilde{h}_i(Q(t)) F(t), \qquad i \in \{j, a\}. \tag{31b}$$

After dropping the tildes, the nondimensional model is

$$\frac{dx}{dt} = x(t)(1 - x(t))\left(1 - \frac{1}{Q(t)}\right) - F(t)\big(J(t) + A(t)\big), \tag{32a}$$

$$\frac{dJ}{dt} = \alpha D_j(t)\left[A(t) - A(t-\tau(t))S(t)\right] - \frac{\chi_j}{D_j(t)}J(t), \tag{32b}$$

$$\frac{dA}{dt} = \beta D_j(t)A(t-\tau(t))S(t) - \frac{\chi_a}{D_a(t)}A(t), \tag{32c}$$

$$1 = \rho\int_{t-\tau(t)}^{t} D_j(\xi)\,d\xi, \tag{32d}$$

$$S(t) = \exp\left\{-\chi_j\int_{t-\tau(t)}^{t}\frac{1}{D_j(\xi)}\,d\xi\right\}. \tag{32e}$$

11.4 Change of Variables

Now we perform a change of variables from chronological time t to developmental time ϕ, using the relation

$$\phi(t) = \rho\int_0^t D_j(\xi)\,d\xi, \qquad \frac{d\phi(t)}{dt} = \rho D_j(t). \tag{33}$$

Note that we can invert this relation, yielding

$$\frac{dt}{d\phi} = \frac{1}{\rho D_j(\phi)}. \tag{34}$$

Therefore, the complete transformed system is

$$\frac{dx(\phi)}{d\phi} = \frac{x(\phi)}{\rho D_j(\phi)}\left(1 - x(\phi)\right)\left(1 - \frac{1}{Q(\phi)}\right) - \frac{J(\phi) + A(\phi)}{\rho h_j(Q(\phi))} \tag{35a}$$

$$\frac{dJ(\phi)}{d\phi} = \frac{\alpha}{\rho}\left[A(\phi) - A(\phi-1)S(\phi)\right] - \frac{\chi_j}{\rho D_j(\phi)^2}J(\phi) \tag{35b}$$

$$\frac{dA(\phi)}{d\phi} = \frac{\beta}{\rho}A(\phi-1)S(\phi) - \frac{\chi_a}{\rho D_j(\phi)D_a(\phi)}A(\phi) \tag{35c}$$

$$\frac{dt(\phi)}{d\phi} = \frac{1}{\rho D_j(\phi)} \tag{35d}$$

$$S(\phi) = \exp\left\{-\frac{\chi_j}{\rho}\int_{\phi-1}^{\phi}\frac{1}{D_j(\psi)^2}\,d\psi\right\}. \tag{35e}$$

Here, $t(\phi)$ records nondimensional chronological time. The chronological stage duration in days can be recovered using

$$\tau_{\text{days}}(\phi) = \frac{t(\phi) - t(\phi-1)}{b}. \tag{36}$$

11.5 Survival Probability Methods

Notice that our survival probability

$$S(\phi) = \exp\left\{-\frac{\chi_j}{\rho}\int_{\phi-1}^{\phi}\frac{1}{D_j(\psi)^2}\,d\psi\right\}. \tag{37}$$

is a state-dependent integro-differential equation. And thus we must approximate this probability if we hope to use BIFTOOL for bifurcation analysis. However, for normal numerical simulation through `dde23`, we can use the following method. Firstly, define an additional differential equation as

$$\frac{dR}{d\phi} = \frac{1}{D_j(\phi)^2} \tag{38}$$

and by the fundamental theorem of calculus

$$R(\phi) - R(\phi-1) = \int_{\phi-1}^{\phi}\frac{1}{D_j(\psi)^2}\,d\psi. \tag{39}$$

Therefore, for normal numerical simulations we can solve for the survival probability using

$$S(\phi) = \exp\left\{-\frac{\chi_j}{\rho}[R(\phi) - R(\phi-1)]\right\}. \tag{40}$$

However, for bifurcation analysis, we must use the method outlined in the McCauley Supplement and further explained in our write-up of that model. We will leave the more detailed explanation in that document and only present the final approximation.

Firstly, we define

$$z(\phi) = \frac{1}{\zeta}\frac{1}{D_j(\phi)^2} \tag{41}$$

and thus the integrand inside $S(\phi)$ can be rewritten to be ,

$$\int_{\phi-1}^{\phi}\frac{1}{D_j(\psi)^2}\,d\psi = \zeta\int_{\phi-1}^{\phi} z(\psi)\,d\psi. \tag{42}$$

We will now approximate $\int_{\phi-1}^{\phi} z(\psi)\,d\psi$ using a series of low-pass filters defined as such:

$$\frac{dL_i(\phi)}{d\phi} = \begin{cases} -\nu L_i(\phi) + z(\phi), & \text{if } i = 1 \\ -\nu L_i(\phi) + L_{i-1}(\phi), & \text{if } i > 1 \end{cases}$$

Finally, our survival probability will be approximated as

$$S(\phi) \approx \exp\left[-\zeta\frac{\chi_J}{\rho}\left(\sum_{i=1}^{n}\nu^{i-1}\left[L_i(\phi) - L_i(\phi-1)\right] + \nu^n L_n(\phi)\right)\right]$$

Note that McCauley found that $\nu = 1$, $n = 5$, and $\zeta = 1000$ were sufficient to approximate $S(\phi)$.

12 VERY Preliminary Parameters, Code, Figures, and Discussion

In this section, we will outline some of the parameter decision, code, and initial simulation behaviors from our variable-mortality combined model.

12.1 Parameters

To start, let us discuss our new parameters and some potential values. In order for our variable death terms $\delta_j(t)$ and $\delta_a(t)$ to have units of $\frac{1}{\text{days}}$, we must introduce mortality scalers μ_j and μ_a which have units of $\frac{1}{\text{days}^2}$. These units differ from their corresponding parameters in the McCauley model seen in Table 2. It is important to note that in the McCauley simulations, their mortality scalers μ_i have extremely small values[2] leading towards low mortality, and most importantly high through stage survival probability. We are unsure of how to calibrate this parameters, but found that in order see larger stage duration than cycle period, we needed to keep them quite small. However, if we chose parameter values that were too small, we failed to see an eventual collapse in the bifurcation diagram for higher K values. We think this is due to the poor food quality becoming insignificant within the death term and through stage survival probability.

12.2 Code

All code used in this section will be included in the GITHUB repository linked at the top of the file. Specifically, the figures in this section were made with `combined_model_variable_death_simulations_K.m` which uses the first survival probability method outline in section 11.5. As a potentially helpful reference, we also include `approximated_survival_code.m` which outlines the history function and dde function which would be used for bifurcation analysis.

While it is not difficult to replicate the varying w and K, bifurcation, and heatmap figures seen the above sections with this variable death system, we found that the solving of this system through dde23 takes significantly longer than the our constant mortality system. Thus, analysis is more limited by computation power/time constraints.

12.3 Initial Population Dynamics

In this section, we will show some population dynamic figures as well as the stage duration and cycle period over time. We use the same parameter values as seen in Table 5, except for the constant death rates which are no longer relevant, and mortality scalers of $\mu_j = 1.09 \cdot 10^{-5}, \mu_a = 5.92 \cdot 10^{-6}$. We also noticed that in our simulations, higher values of w led to longer stage durations and with our variable death model, we were able to see stage duration's longer than cycle

[2] $\mu_j = 1.09 \cdot 10^{-5}, \mu_a = 5.92 \cdot 10^{-6}$

oscillations. In Figure 14, we show the population dynamics for $K = 1.5, w = 2$ and corresponding stage duration in Figure 15. For all figures in this section, we use a constant history function of $x(s) = 1.08, J(s) = 0.04, A(s) = 0.04$ and a linear history function $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$. We chose this since McCauley often references an resource-rich history.

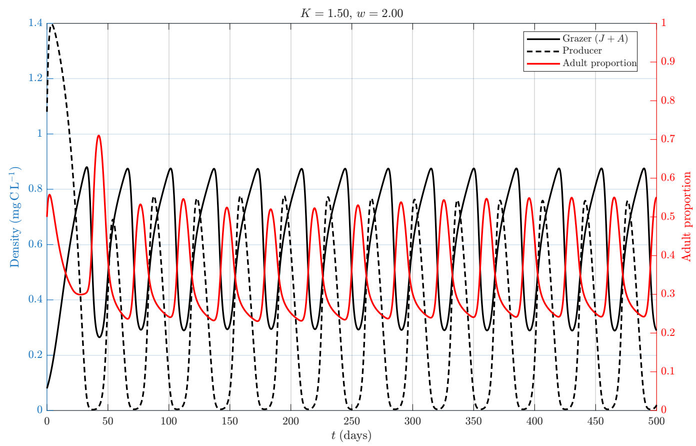


Figure 14: Population dynamics from our variable death combined model with $K = 1$ and $w = 2$.

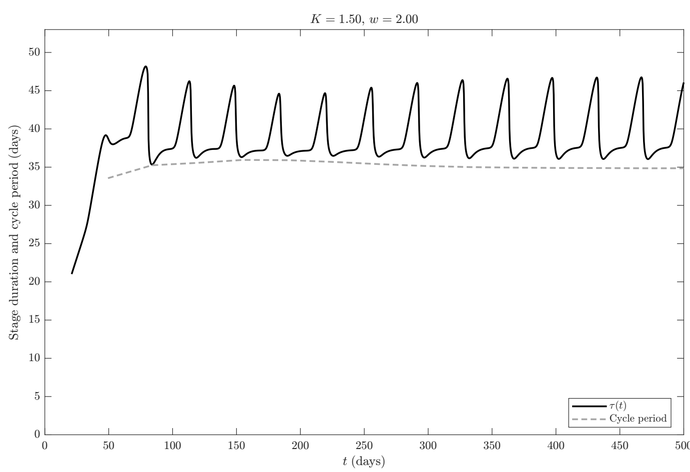


Figure 15: Stage duration and cycle period plot from our variable death combined model with $K = 1$ and $w = 2$.

Important to note that with this model, we were finally able to observe stage duration's longer than cycle periods. This specific parameter combination was unable to yield a switch in average stage duration but we believe it to be possible given the right w, K and initial conditions. It also might be interesting to note that although the producer line $J + A$ does not have a noticeable change in amplitude early on, the Adult proportion does go from a large period cycle to a smaller period cycle.

Below in Figure 16 we show the stage duration plot for $K = 1.5$ and $w = 1$ to showcase that for lower values of w, we see stage duration's below cycle periods.

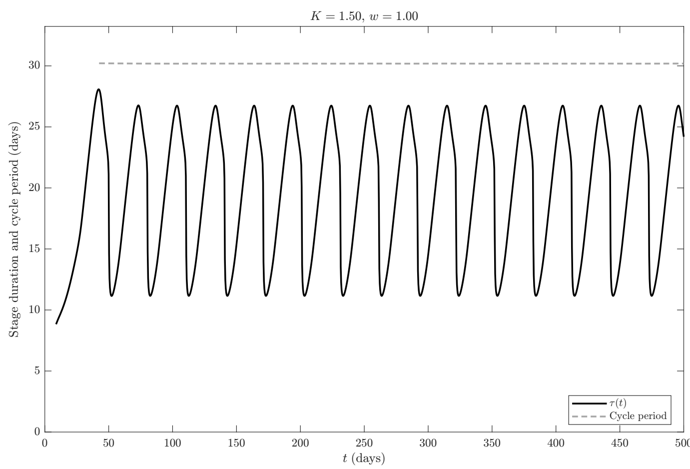


Figure 16: Stage duration and cycle period plot from our variable death combined model with $K = 1$ and $w = 1$.

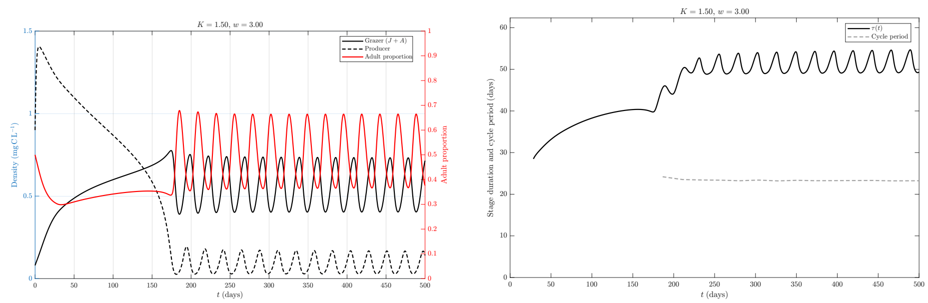
12.4 More Population Dynamics and Bifurcation Diagrams

In this section we will present more population dynamics as well as some bifurcation diagrams. Through some brute force exploration using AI written code, we found a combination of parameter values that yielded both a stage duration higher than cycle period and an eventual collapse to stable equilibrium for higher K values. If you, the reader, is curious for the code or prompts that yielded this parameter combination email `aharden@haverford.edu` but the general idea was to brute force test a variety of mortality scalers, w values, and K values which yielded the desired behavior [3]. Through our exploration, we landed on

$$\mu_j = 7.45 \cdot 10^{-3}, \quad \mu_a = 5.92 \cdot 10^{-6}, \quad w = 3, \quad K = 1.5$$

which can be seen in Figure 17. For all figures in this section, we use a constant history function of $x(s) = 0.9, J(s) = 0.04, A(s) = 0.04$ and a linear history function $t(s) = \frac{1}{\rho D(s)}$ for all $s < 0$.

[3] Some discussion of this process was summarized by ChatGPT in `numerical_search_addendum.pdf`



(a) Population dynamics.

(b) Stage duration and grazer cycle period.

Figure 17: Population dynamics and stage-timing behavior for $K = 1.5\ \mathrm{mg\,C\,L^{-1}}$ and $w = 3$.

Notice that in these figure, there is a buffering period before the cycles begin. We will discuss this in a later section, but this indicates that we could be using custom history functions to yield this cycle behavior starting at $t = 0$. From McCauley, it seemed that they used a similar method to yield certain figures.

Now we will present a bifurcation diagram with the same parameters in Figure 18 that showcases the desired behavior cycle collapse from Ascoli. The code used to generate the Figure 18 is under `variable_mortality_bifurcation_scan.m`

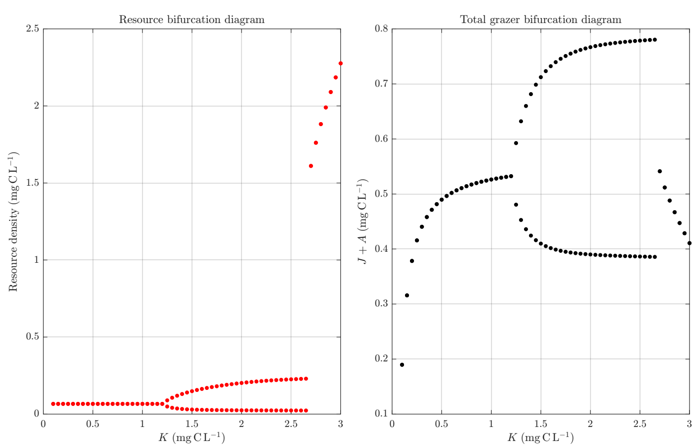


Figure 18: Diagram showcasing long term maximum and minimum for x and $J + A$

Thus we know it is possible to have longe stage duration, and still collapse to interior equilibrium at larger K.

12.5 Comments on History Functions

For all simulations in this document, we have used a constant population history and a linear time history. This is largely done for simplicity sake but as seen in Figure 17, we can see a buffering period before the cycles begin. Thus a potential strategy for seeing cycles appear at $t = 0$ would be to first solve with constant histories, then grab the output for $\phi = 1$ units before the cycles begin, and use that as a history function for the next solve. We briefly tried this method with AI written code and it yielded cycles beginning at $t = 0$.

13 Final Comments

In this section we will make some closing comments on both model's presented in this document and discuss some next steps.

Let us first discuss the constant mortality combined model. We were quite pleased with the results of these model and its ability to combine the structure of Ascoli and McCauley. The difference in units of J and A proved to be a consequential difference, requiring a dimensionless w term, and therefore a new parameter in need of tuning. However, it was possible to incorporate ecological stoichiometry into the delayed system of equation proposed in McCauley. We found mixed results in reproducing all of Ascoli's and McCauley's behavior but found success in some aspects, such as bifurcation diagrams matching the LKE model. We were unable to found small and large amplitude limit cycles with this model as well as abrupt changes in stage duration. We were also unable to see stage duration longer than cycle period. This could be due to lack of initial conditions testing, incorrect parameters, or the lack of resource dependent mortality.

Now let us discuss the variable mortality combined model. Again the differing units required the addition of new mortality scalers μ_j and μ_a which we found to have a significant impact on behavior. We noticed that these scalers needed to be small enough to yield long stage duration, but large enough that mortality/survival probability was effected by poor food quality. We were able to see the same bifurcation structure as in Ascoli and a stage duration longer than cycle period. We were still unable to find large and small amplitude cycles but feel that it might be possible with the correct histories (likely not constant) and parameter combinations.

Based on our work in this document, we feel that further exploration should be focused on the variable death model as it allowed for a wider variety of stage duration behavior. Testing a range of mortality scalers and w values to identify which combinations yield the correct bifurcation structure and stage duration would be a good next step. Additionally, we want to be able to use the BIFTOOL framework to hopefully match Figure 1 in McCauley in identifying stable and unstable limit cycles.